
TECHWORLD ELECTRONICS

Data Analytics Report

Phase 2 Mini-Project

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Database: techworld_db

Tool: PostgreSQL / pgAdmin

Period: January 2023 – December 2025

1. Executive Summary

This report presents the findings from a comprehensive SQL-based analysis of TechWorld Electronics' transactional database, covering three years of commercial activity from January 2023 through December 2025. The analysis was conducted using PostgreSQL and pgAdmin against the techworld_db database, which contains 80 customers, 40 products, 25 employees, 200 orders, and 530 order line items across five interconnected tables.

Seven analytical problems were solved to generate actionable business intelligence across four key domains: customer value, product performance, sales operations, and workforce productivity.

Key Findings at a Glance

Total Revenue Analysed:	\$317,508.98 across all valid orders
Top Revenue Category:	Laptops at \$109,504.70 (34.5% of total)
Best Sales Channel:	Phone at \$103,067.36 with lowest return rate (5.7%)
Top Employee:	Luisa Fernandez (Europe) - \$4,372.00 commission earned
Highest Return Category:	Monitors at 16.7% return rate
Fastest Growing Month:	December 2024 - +538.2% YoY growth
Most Efficient Segment:	Regular customers at \$4,771.29 revenue per customer

2. Database Overview

The techworld_db database was set up in pgAdmin by executing the provided SQL script. The following verification confirmed all tables were successfully created and populated:

Table	Records	Description
customers	80	Demographics, segmentation, loyalty, preferred channel
products	40	40 SKUs across 10 categories and 3 brands
employees	25	25 staff across 6 departments with commission structures
orders	200	Full lifecycle order tracking across 3 years
order_items	530	Individual line items per order with pricing and discounts

Five pre-built views were also created to accelerate analysis: vw_customer_lifetime_value, vw_product_performance, vw_monthly_sales_trend, vw_employee_sales_performance, and vw_regional_sales. These views handle complex joins and pre-filter cancelled and returned orders, forming the foundation for most analytical queries.

3. Analytical Findings

3.1 Top 10 Customers by Lifetime Revenue

Using the vw_customer_lifetime_value view, customers were ranked by total spend across all valid (non-cancelled, non-returned) orders. The view joins the customers and orders tables, computes lifetime revenue, average order value, and order count per customer.

Customer	Segment	Country	Orders	Lifetime Revenue	Avg Order Value
John Doe	Enterprise	United States	-	Highest	High
Hiroshi Suzuki	Enterprise	Japan	-	2nd highest	High
Michael Thompson	Enterprise	United States	-	3rd highest	High
Thabo Molefe	Enterprise	South Africa	-	Top 10	High
Wei Zhang	Enterprise	China	-	Top 10	High
Jennifer Anderson	Enterprise	United States	-	Top 10	Medium
Hans Schmidt	Enterprise	Germany	-	Top 10	High
James Wilson	Enterprise	United Kingdom	-	Top 10	High
Omar Al-Fayed	Enterprise	UAE	-	Top 10	High
Suki Park	Enterprise	South Korea	-	Top 10	Medium

Key Insight: The top 10 is dominated by Enterprise segment customers from high-purchasing-power regions - US, Japan, Germany, UAE, and South Africa. Retaining these customers should be TechWorld's highest priority, as losing even one represents a significant revenue risk.

3.2 Product Category Revenue and Margin Analysis

Using the vw_product_performance view, product-level data was aggregated to category level. Both a simple average margin and a sales-volume-weighted margin were calculated to give a more accurate picture of true profitability.

Category	Products	Units Sold	Revenue	Avg Margin %	Weighted Margin %
Laptops	5	110	\$109,504.70	56.5%	56.6%
Monitors	3	59	\$47,269.41	59.6%	59.6%
Smartphones	5	91	\$47,028.89	56.9%	57.3%
Tablets	4	78	\$39,551.12	57.9%	57.9%
Wearables	3	75	\$20,711.25	66.9%	67.6%
Audio	5	88	\$17,556.32	67.0%	66.9%
Gaming	2	43	\$12,243.97	54.2%	46.2%
Smart Home	4	85	\$11,306.65	68.0%	67.5%
Storage	3	48	\$6,080.02	58.2%	58.1%
Accessories	6	129	\$5,447.51	72.7%	69.7%

Key Insight: Laptops generate more than double the revenue of any other category (\$109,504) but Accessories deliver the highest margin efficiency (69.7%). Gaming has the widest gap between simple and weighted margin (54.2% vs 46.2%), revealing that lower-margin gaming products sold more units - a warning sign for profitability.

3.3 Monthly Revenue Trend with Year-over-Year Growth

A CTE was used alongside the LAG() window function to compare each month's revenue against the same month in the prior year. This technique partitions data by month number and orders by year to compute accurate year-over-year growth percentages.

Period	Notable Month	Revenue	YoY Growth	Observation
2023	May 2023	\$38,066.16	Baseline (NULL)	Highest month in 2023 - 15 orders
2023	Sep 2023	\$3,484.07	Baseline (NULL)	Weakest month in 2023 - 5 orders
2024	Jun 2024	\$22,721.17	+91.7%	Strong Q2 recovery
2024	Oct 2024	\$12,662.34	+161.0%	Holiday season build-up
2024	Dec 2024	\$40,889.41	+538.2%	Best single month in dataset
2025	Nov 2025	\$350.15	-98.4%	Sparse data - only 2 orders
2025	Jul 2025	\$10,122.80	+44.2%	One of few positive 2025 months

Key Insight: December is TechWorld's most critical trading month - December 2024 generated \$40,889 vs \$6,406 in December 2023 (+538%). The 2025 data appears sparse (many months show only 1-2 orders), which artificially depresses YoY figures for that year.

3.4 Sales Channel Effectiveness

The orders table was queried directly using FILTER (WHERE ...) aggregates to calculate revenue, average order value, and cancellation/return rates per channel - all in a single query pass.

Channel	Total Orders	Completed	Revenue	Avg Order Value	Cancel Rate	Return Rate
Phone	53	46	\$103,067.36	\$2,240.59	7.5%	5.7%
Online	55	48	\$80,061.52	\$1,667.95	7.3%	5.5%
In-Store	48	33	\$71,911.86	\$2,179.15	18.8%	12.5%
Mobile App	44	31	\$62,468.24	\$2,015.10	18.2%	11.4%

Key Insight: Phone leads in revenue (\$103,067) and has the lowest problem rate (13.2%

combined). In-Store has a dangerously high cancellation rate (18.8%) despite customers physically inspecting products - this requires urgent operational review.

Online is the most reliable volume channel with the most total orders (55).

3.5 Employee Commission Report

The vw_employee_sales_performance view was queried with a RANK() window function partitioned by region to rank each sales employee within their territory. Results are ordered by estimated commission earned.

Employee	Title	Region	Orders	Total Sales	Rate	Commission	Regional Rank
Luisa Fernandez	Senior Sales Rep	Europe	12	\$36,433.32	12%	\$4,372.00	1
Maria Garcia	Sales Representative	Latin America	20	\$35,248.97	10%	\$3,524.90	1
Priya Sharma	Senior Sales Rep	Asia Pacific	17	\$27,192.70	12%	\$3,263.12	1
Emily Carter	Senior Sales Rep	North America	8	\$25,378.55	12%	\$3,045.43	2
David Johnson	Sales Representative	North America	13	\$29,435.00	10%	\$2,943.50	1
Oluwaseun Adeyemi	Sales Representative	West Africa	10	\$26,939.40	10%	\$2,693.94	1
Sarah Mitchell	Regional Sales Mgr	North America	10	\$25,203.44	8%	\$2,016.28	3
Amina Bello	Regional Sales Mgr	West Africa	15	\$24,170.74	8%	\$1,933.66	2
James Okonkwo	VP of Sales	West Africa	0	\$0.00	0%	\$0.00	4

Key Insight: Luisa Fernandez is TechWorld's top earner with the highest avg order value (\$3,036) - quality over quantity. Maria Garcia handles the most orders (20) and is the sole rep covering Latin America - a business continuity risk. Sales reps are outperforming

their managers in every region, which is expected as managers handle strategy and coaching.

3.6 Return Rate by Product Category

Three raw tables (orders, order_items, products) were joined directly to calculate total units ordered versus units returned per category. This approach was necessary because no single view contained all the required fields in the required combination.

Category	Orders	Units Ordered	Units Returned	Return Rate	Return Orders
Gaming	24	50	0	0.0% (NULL)	0
Monitors	36	78	13	16.7%	4
Audio	55	123	18	14.6%	6
Storage	37	69	10	14.5%	5
Laptops	61	137	19	13.9%	7
Smartphones	57	136	15	11.0%	6
Tablets	50	100	11	11.0%	5
Accessories	76	168	9	5.4%	4
Wearables	43	95	3	3.2%	1
Smart Home	48	100	1	1.0%	1

Key Insight: Monitors have the highest return rate (16.7%) on the 2nd highest revenue category - a costly combination. Laptops have the most absolute returns (19 units / 7 orders) at an avg price of ~\$1,000 each. Smart Home and Wearables are bright spots with near-zero return rates. Gaming recorded zero returns - the strongest product satisfaction.

3.7 Customer Segmentation Revenue Contribution

The vw_regional_sales view was aggregated using two CTEs - one to sum revenue by segment, and one to calculate the grand total. A CROSS JOIN attached the single grand-total row to all segment rows, enabling revenue share percentage calculations.

Segment	Customers	Orders	Revenue	Avg Order Value	Revenue Share	Revenue/Customer
Premium	32	54	\$110,684.65	\$1,874.91	34.9%	\$3,458.90
Regular	23	49	\$109,739.72	\$2,092.63	34.6%	\$4,771.29
Enterprise	25	55	\$97,084.61	\$1,475.79	30.6%	\$3,883.38

Key Insight: The results are counterintuitive - Regular customers generate the highest avg order value (\$2,092) and revenue per customer (\$4,771), suggesting they make high-value one-off purchases. Revenue is remarkably balanced across all three segments (~33% each), indicating TechWorld has not over-indexed on any single tier.

4. Business Recommendations

4.1 Customer Strategy

- Implement a dedicated account management programme for the top 10 Enterprise customers - they represent a disproportionate share of lifetime value.
- Launch a Regular-to-Premium upgrade programme - Regular customers already spend \$4,771 per head, more than Premium customers. Converting them would unlock higher loyalty rewards engagement.
- Invest in Premium customer retention - they contribute the most total revenue (34.9%) and have the most customers (32). Churn prevention here has the highest financial impact.

4.2 Product Strategy

- Bundle accessories with laptop and smartphone purchases - Accessories have the highest margin (69.7%) but the lowest revenue (\$5,447). Cross-selling them with high-volume categories is a quick margin win.
- Investigate Monitor product quality and descriptions - a 16.7% return rate on a \$47,000 revenue category is unsustainable. Improved compatibility guides and quality control are recommended.
- Study Gaming's zero-return performance - understanding what makes gaming customers satisfied could offer lessons applicable to higher-return categories like Monitors and Audio.

4.3 Sales Channel Strategy

- Invest in the Phone channel - it generates the most revenue (\$103,067) with the lowest problem rates. Hiring additional phone sales reps would likely deliver strong ROI.
 - Urgently review In-Store operations - an 18.8% cancellation rate and 12.5% return rate at physical stores suggests issues with staff training, inventory accuracy, or pressure selling tactics.
 - Improve the Mobile App experience - a combined 29.6% cancellation and return rate suggests the app may be misleading customers or enabling accidental orders. A UX audit is recommended.
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4.4 Workforce Strategy

- Recognise and retain Luisa Fernandez - top commission earner with the highest avg order value in the company. She is a flight risk if not compensated competitively.
- Hire a second sales representative for Latin America - Maria Garcia is the sole coverage for the entire region. Her departure would leave a significant revenue gap.
- Review commission structure fairness - Sales Representatives are outperforming Regional Sales Managers in revenue generation. Consider performance-based commission tier upgrades.

4.5 Seasonal Strategy

- Maximise Q4 readiness - December is clearly TechWorld's peak month. Inventory build-up, targeted marketing campaigns, and additional staffing should begin no later than September each year.
- Stimulate Q1 and Q3 - these are historically lighter quarters. Promotions, bundle deals, and channel-specific campaigns could help smooth revenue seasonality.

5. Conclusion

This analysis demonstrates the power of structured SQL analytics in generating actionable business intelligence from transactional data. Using a combination of pre-built views, raw table joins, window functions (LAG, RANK), CTEs, and aggregate filtering, seven distinct business questions were answered from a single PostgreSQL database.

The most significant findings are that TechWorld's Phone channel is underappreciated, its In-Store channel has operational problems, Monitors carry an unsustainable return rate, and Regular customers are generating surprisingly high per-customer revenue. Addressing these four areas alone could materially improve both revenue and margin performance.

The dataset, while synthetic and limited to 200 orders, is structurally sound and representative of real-world commercial complexity. The analytical techniques applied here scale directly to production-scale databases and form a strong foundation for more advanced analytics including cohort analysis, predictive churn modelling, and real-time dashboard development.
